

Logistic Regression

Fitted and Residuals

In this lecture

- Fitted values
- Residuals
- Diagnostics

Notation

To simplify the notation, let L_i be the linear component and p_i the conditional probability:

$$L_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_p X_{pi}$$

and,

$$E \left[Y_i \mid X_{1i}, \dots, X_{pi} \right] = P \left[Y_i = 1 \mid X_{1i}, \dots, X_{pi} \right] = p_i$$

Logistic Regression: equivalent forms

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Recall that a logistic model can be written in different forms.

$$p_i = \frac{e^{L_i}}{1 + e^{L_i}} \quad [\text{probability}]$$

$$\underbrace{\log\left(\frac{p_i}{1 - p_i}\right)}_{\text{the logit function}} = L_i \quad [\text{log-odds}]$$

$$\underbrace{\left(\frac{p_i}{1 - p_i}\right)}_{\text{odds}} = e_i^L \quad [\text{odds}]$$

Prediction

The estimated model can be used to predict an outcome

- the predicted outcome depends on the form of the model
 - log-odds (default, since this is a linear model)
 - probabilities
 - odds
- we can predict the outcome of:
 - observations in the same sample used to estimate the model, called **fitted** values
 - new observations, usually referred as **predicted** values

Dataset and model

Let's use the [Titanic dataset](#) and an additive model with [sex](#) and [fare](#) as covariates to illustrate concepts.

```
1 model_titanic_logistic_additive <- glm(survived ~ sex + fare, data = titan,  
2                                     family = binomial)  
3  
4 tidy(model_titanic_logistic_additive) %>%  
5   mutate_if(is.numeric, round, 3) %>%  
6   knitr::kable()
```

term	estimate	std.error	statistic	p.value
(Intercept)	0.647	0.149	4.358	0
sexmale	-2.423	0.171	-14.208	0
fare	0.011	0.002	4.886	0

Prediction

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Suppose we want to use the predicted model to predict the odds of a male passenger who paid a fare of \$7.25.

The predicted *log-odds* are

$$\log\left(\frac{\hat{p}_s}{1 - \hat{p}_s}\right) = \underbrace{0.647}_{\hat{\beta}_0} - \underbrace{2.423}_{\hat{\beta}_1} \times 1 + \underbrace{0.011}_{\hat{\beta}_2} \times 7.25 = -1$$

Next, by exponentiating both sides of the equation, we obtain

$$\frac{\hat{p}_s}{1 - \hat{p}_s} = e^{-1.696} = 0.183$$

Finally, solving the above for $\hat{p}_s$, we obtain our predicted probability of surviving

$$\hat{p}_s = \frac{e^{-1.696}}{1 + e^{-1.696}} = \frac{0.183}{1.183} = 0.155$$

in R

predicted log-odds (default)

```
1 predict(model_titanic_logistic_additive,  
2         tibble(sex = "male", fare = 7.25),  
3         type = "link") %>% round(3)
```

```
1  
-1.694
```

predicted probabilities

```
1 predict(model_titanic_logistic_additive,  
2         tibble(sex = "male", fare = 7.25),  
3         type = "response") %>% round(3)
```

```
1  
0.155
```


Fitted values

In R there are different ways of obtaining *fitted* values.

It is important to know which forms are given by which function.

- the `augment()` function adds `.fitted` to the dataset: these are predicted *log-odds*
- the element `$fitted` or function `fitted()` of the estimated model object gives the predicted *probabilities*
- predicted *odds* can be computed from these columns

```

1 model_titanic_logistic_additive %>%
2   augment() %>%
3   dplyr::select(survived,sex, fare,.fitted) %>%
4   mutate(pred_prob1 = model_titanic_logistic_additive$fitted,
5          pred_prob2 = fitted(model_titanic_logistic_additive))
6   head() %>%
7   mutate_if(is.numeric, round, 3) %>%
8   knitr::kable()

```

survived	sex	fare	.fitted	pred_prob1	pred_prob2
0	male	7.250	-1.694	0.155	0.155
1	female	71.283	1.446	0.809	0.809
1	female	7.925	0.736	0.676	0.676
1	female	53.100	1.243	0.776	0.776
0	male	8.050	-1.685	0.156	0.156
0	male	8.458	-1.681	0.157	0.157

Residuals

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The **raw residuals** are, as usual, the differences between the observed and the fitted values.

$$\begin{aligned} r_i &= \text{obs}_i - \text{fit}_i \\ &= y_i - \hat{p}_i \end{aligned}$$

However, there are 2 problems with these residuals:

- the residuals are not properly scaled since the variance of the observations is not constant! for a binary response:

$$\text{Var}(Y_i) = p_i(1 - p_i)$$

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technically, we should write the conditional variance and probability

Thus, residuals of different observations are not comparable and should be adjusted!!

- the raw residuals have values $-\hat{p}_i$ and $1 - \hat{p}_i$, respectively, since the y_i take only 2 values $y_i = 0$ and $y_i = 1$. This creates 2 lines in usual residual plots.

Residuals for GLMs

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Many variations of the raw residuals have been proposed for GLMs that take the variance of the observations into account.

- **Pearson residuals:** divides the raw residuals by the standard deviation of the response

$$r_i = \frac{y_i - \hat{p}_i}{\sqrt{\hat{p}_i(1 - \hat{p}_i)}}$$

- **deviance residuals**
- **standardized residuals:** for both Pearson and deviance residual

We won't get to the details of these but it is important to know which ones you are calculating

```

1 model_titanic_logistic_additive %>%
2     augment() %>%
3     dplyr::select(survived,sex, fare,.fitted, .resid,.std.resid)
4     mutate(pred_prob = model_titanic_logistic_additive$fitted,
5             resid_raw = residuals(model_titanic_logistic_additive,
6                                   raw_byhand = survived - pred_prob,
7                                   resid_deviance = residuals(model_titanic_logistic_addi
8                                   resid_pearson =residuals(model_titanic_logistic_additi
9                                   pearson_byhand = resid_raw/sqrt(pred_prob*(1-pred_prob)
10                                resid_standardized =rstandard(model_titanic_logistic_a
11     head() %>%
12     mutate_if(is.numeric, round, 3) %>%
13     head(3) %>%
14     knitr::kable()

```

survived	sex	fare	.fitted	.resid	.std.resid	pred_prob	resid_raw	raw_by
0	male	7.250	-1.694	-0.581	-0.581	0.155	-0.155	-0.15
1	female	71.283	1.446	0.650	0.652	0.809	0.191	0.19
1	female	7.925	0.736	0.885	0.887	0.676	0.324	0.32

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```

model_titanic_logistic_additive %>%
  aknitr::kable
  dplyr::select(survived,sex, fare,.fitted, .resid,.std.resid) %>%
  mutate(pred_prob = model_titanic_logistic_additive$fitted,
         resid_raw = residuals(model_titanic_logistic_additive, type= "response"),
         raw_byhand = survived - pred_prob,
         resid_deviance = residuals(model_titanic_logistic_additive),
         resid_pearson =residuals(model_titanic_logistic_additive,"pearson"),
         pearson_byhand = resid_raw/sqrt(pred_prob*(1-pred_prob)),
         resid_standardized =rstandard(model_titanic_logistic_additive)) %>%
  head() %>%
  mutate_if(is.numeric, round, 3) %>%
  head(3) %>%
  knitr::kable()

```

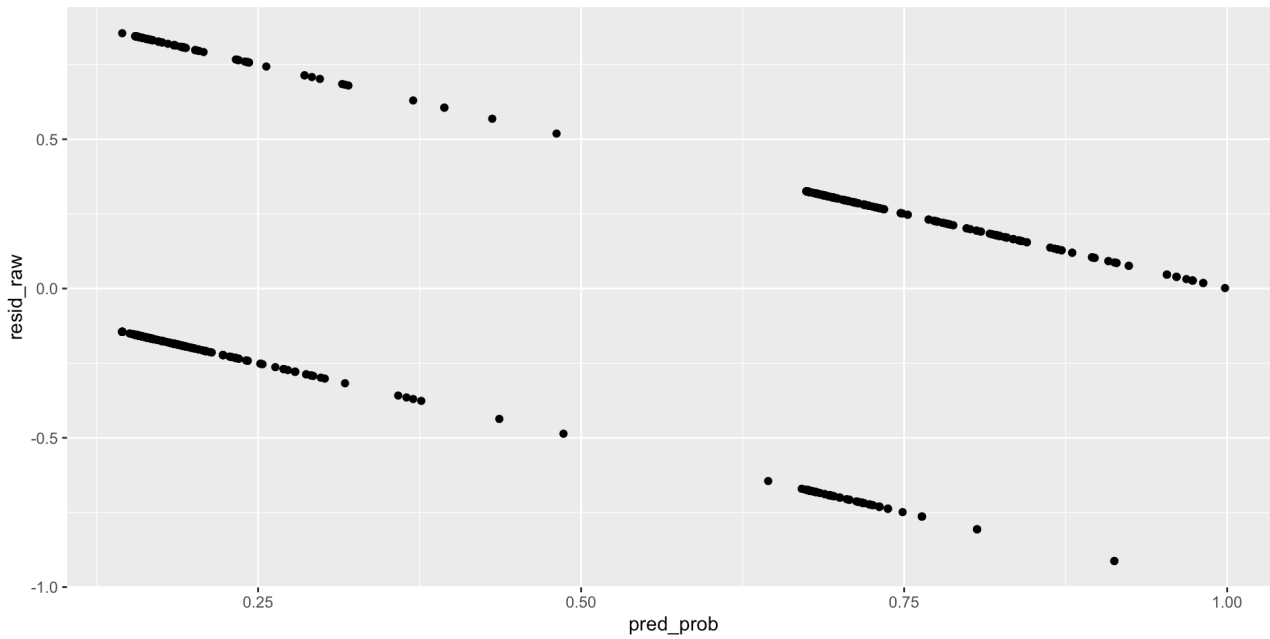
Diagnostics

As we mentioned before, the raw residuals give only 2 values $-p_i$ or $1 - p_i$, which creates 2 straight lines in the residuals plots

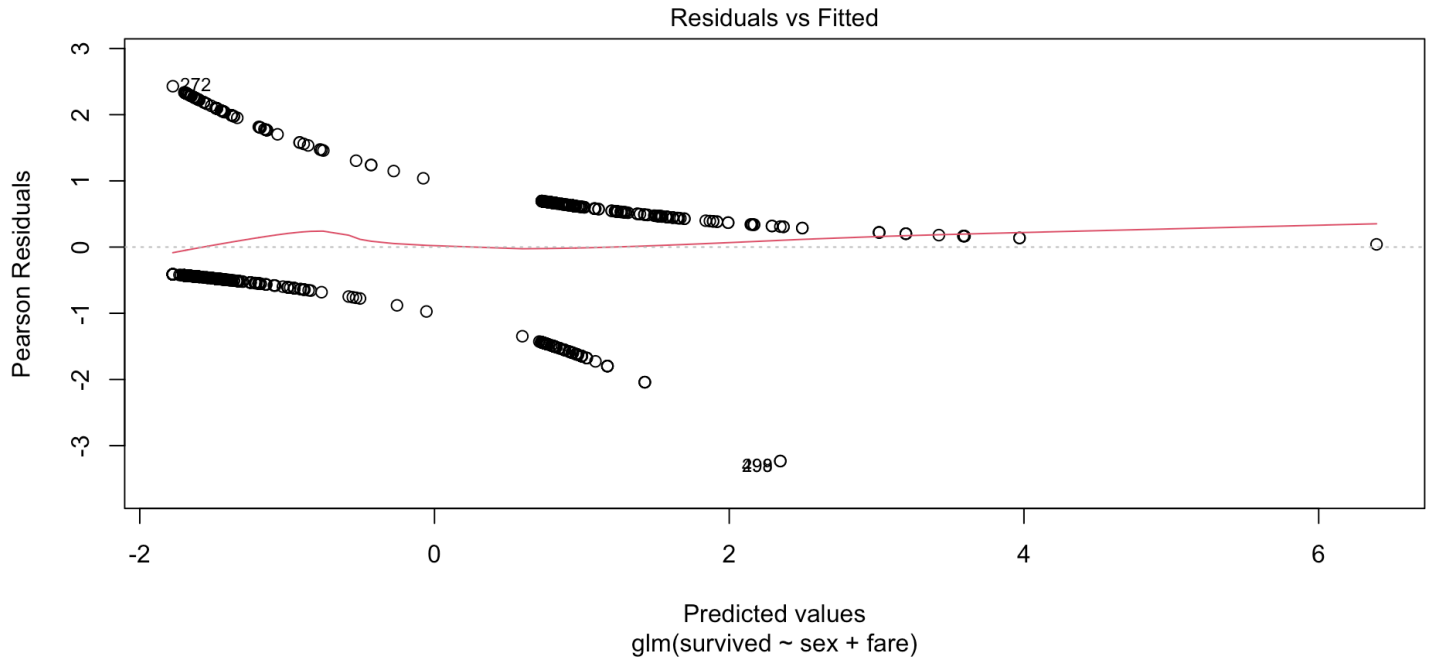
Not much is gained using the Pearsons residuals, the problem stems from having a binary response

Although these are hard to interpret, we don't want to see clear patterns in a residuals plot.

```
1 model_titanic_logistic_additive %>%  
2   augment() %>%  
3   mutate(pred_prob = model_titanic_logistic_additive$fitted,  
4   ggplot(aes(x= pred_prob,y=resid_raw)) +  
5   geom_point())
```



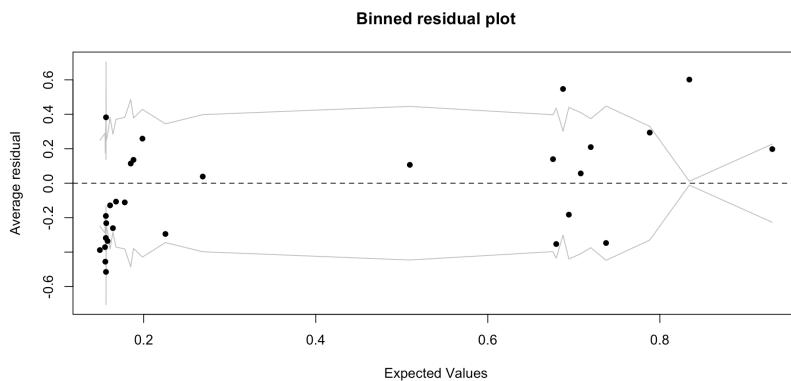
```
1 plot(model_titanic_logistic_additive,1)
```



Binned residuals plot

To address this problem, the binned plot has been proposed. Points are binned and the summary of each bin is plotted.

```
1 y_resid <- residuals(model_titanic_logistic_additive)
2 x_fit <- model_titanic_logistic_additive$fitted
3
4 residual_plot <- binnedplot(x_fit, y_resid)
```



Overdispersion

A more important problem to address in the case of logistic regression is the problem of **overdispersion**.

- As mentioned before, the logistic regression is built assuming that the response is *Bernoulli*, which means that the estimated variance of the response has to be $\hat{p}(1 - \hat{p})$.
- Unfortunately, in real applications, even in situations where the model seems to be estimating the mean well, the data's variability is not quite compatible with the model's assumed variance.

Fixing the overdispersion

This misspecification in the variance affects the SE of the coefficients, not their estimates.

- A more flexible estimate of the model can be obtained using a method called *quasi-maximum likelihood*
- The estimated model also gives an estimate of the **overdispersion parameter**, as well as correct the standard error of our estimators.

_ If the model is correctly specified, the estimated **overdispersion parameter** is 1. Values above or below 1 indicate *over-* or *under-* dispersion

An easy implementation is obtained using the `family` argument to a `quasibinomial`.

```
1 summary(glm(
2   formula = survived ~ sex + fare,
3   data = titan,
4   family = quasibinomial))
```

Call:

```
glm(formula = survived ~ sex + fare, family = quasibinomial,
    data = titan)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.647100	0.146953	4.403	1.20e-05	***
sexmale	-2.422760	0.168736	-14.358	< 2e-16	***
fare	0.011214	0.002271	4.937	9.47e-07	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasibinomial family taken to be 0.9792377)

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Note that the estimated overdispersion parameter (called *Dispersion parameter* in R) is very close to 1 (0.98) indicating no signs of overdispersion.

Although there exists a formal test for overdispersion but we won't cover that in the course.